For NPRE 397, your total grade will be earned through multiple computational exercises as well as contribution to a comprehensive suite of enhancements to the Python for Reactor Kinetics toolkit (PyRK). This independent study is intended to tie together ideas regarding reactor physics, reactor kinetics, data science, reproducible scientific compuing, and numerical methods for nuclear engineering. **This project is more substantial and advanced in nature. It will require 6-10 hours of effort per week and will therefore earn 2 credit hours.**

The primary work will be to demonstrate the PyRK toolkit in the context of an advanced reactor design not yet covered by the suite of example problems. This effort may include resolving open issues in the PyRK repository and implementing improvements to the gaseous density model within. This work will emphasize usability and reproducibility of the toolkit, with special emphasis on demonstrating physics capabilities relevant to high temperature gas reactors (HTGRs). Software development can take place using personal or local workstations, or if needed, cluster computing capabilities. Data and software contributions resulting from this work will be contributed to the open source repository at github.com/pyne/pyrk.

The student will be expected to consistently handle assigned issues in the PyRK repository and to complete a final report and short presentation summarizing their accomplishments by the end of the semester. The project will be assessed as independent research work, much like a journal article undergoes peer review. I will be looking for :

- Relevance
- Technical Detail
- Analytic Rigor
- Verifiability
- Clarity
- A Conclusion

This work will consist of four deliverables:

- a work plan,
- simulation of reactor transients in an HTGR,
- a brief summary report in the style of a conference or journal paper,
- and a 10 minute summary presentation.

1. (20 points) **Work Plan: Due 2025.09.05**

   To help establish scope and milestones, the first step of the project will be a work plan. This plan, developed under supervision of an Advanced Reactors and Fuel Cycles Research Group Graduate Research Assistant, will be delivered in the form of multiple GitHub issues with descriptions and deadlines describing the work you intend to do in the ARFC group. These issues should be compiled into a GitHub Project in the ARFC organization.

   Please arrange at least one short meeting with Prof. Huff to discuss the project plan so that I can help you refine your schedule. I would be happy to provide feedback on a draft of your plan (once) before it is due.

2. (60 points) **Software Contributions: Due 2025.12.10** Complete the full suite of GitHub Issues agreed upon as a part of the Work Plan, within the due dates specified in each issue. All issues must be completed by 2025.12.10. This will require responsive completion of all necessary pull requests in collaboration with colleages within the ARFC group.

3. (20 points) **Final Report and Presentation: Due 2025.12.10** The final report must be submitted to Prof. Huff and a summary presentation will be given to the ARFC group during a weekly group meeting late in the semester.